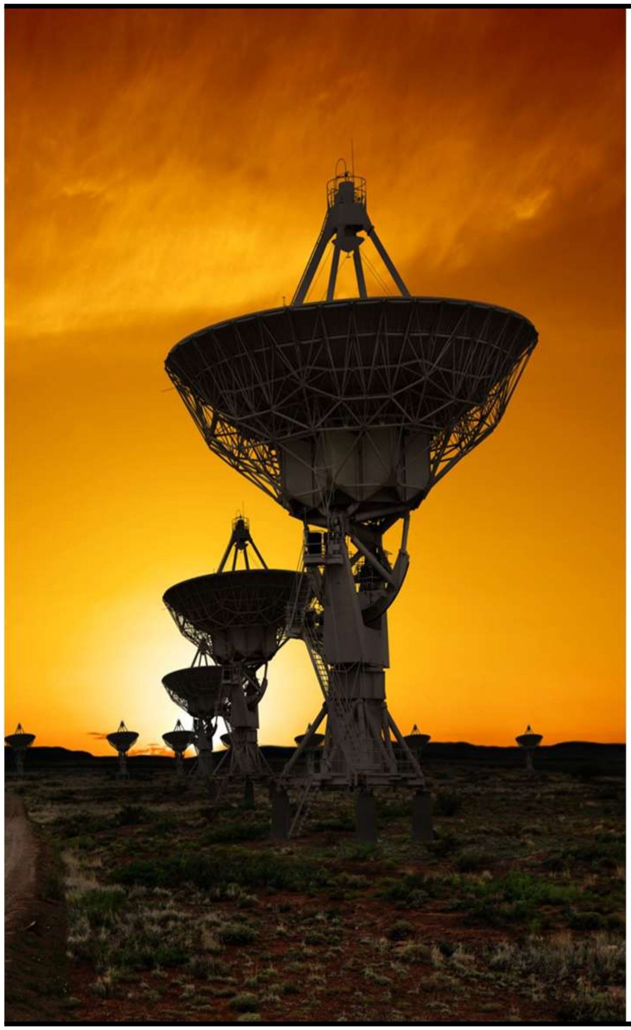


ANALYSIS AND PREDICTION OF WEATHER PATTERNS USING SATELLITE OBSERVATIONS AND MACHINE LEARNING MODELS

SUBJECT NAME: PROJECT STAGE-II
COURSE CODE :EC881
DEPARTMENT: ECE

SEMESTER-8TH
SESSION-2025-26

YEAR: 4TH YEAR

GROUP: 3

Under the Guidance of
Dr. Tuhina Halder

NAME	CLASS ROLL	UNIV ROLL NO
AGNIK GHOSH	29	12200322044
ABHRANIL CHAKRABORTY	09	12200322025
SANTANU MONDAL	37	12200322013
CRASIVA CHAUDHURY	16	12200322050

Problem Statement

Problem

- High-end **NWP models** are computationally expensive and fail to capture localized monsoon volatility in Kolkata

Opportunity

- Big data and advanced machine learning offer modern alternatives to capture intricate, non-linear atmospheric patterns.

Solution

- We leverage **NASA's MERRA-2 and GPM satellite observations** paired with a two-stage Gradient Boosting pipeline

Objectives

- Analyse Local Weather Data
- Feature Pipeline Engineering
- **Dual-Stage GBM Architecture** for predicating weather pattern

Literature Review

Ref No.	Year	Authors	Paper	Research	Gaps
[1]	2024	D. Sharma, P. Rattan	Rainfall Prediction using **Random Forest and XGBoost**-A comparative study	• Compared **Random Forest (RF)** and **XGBoost** using a 10-year NASA dataset (2012–2022) for the Jammu Region. • RF performed slightly better on test data with a lower **RMSE (3.799)**.	• Significant **overfitting** noted (large gap between training and validation error).

Ref No.	Year	Authors	Paper	Research	Gaps
[2]	2024	Jagadeshwari Puttanapura, Monika Singh, C.K.K Reddy, Srinath Doss	Improvised Prediction of Rainfall using Random Forest Classifier Over Taiwan	• Evaluated 9 ML/DL algorithms (including LSTM, SVM, ANN) on a 20-year weather dataset for Taiwan • **Random Forest** outperformed others	• Evaluated rainfall only as a **binary/multi-class classification** target rather than predicting exact quantitative volume (mm/hr).
[3]	2025	A. Jain, G. Sharma	Enhancing Weather Prediction Accuracy with Gradient Boosting: A Machine Learning Approach	• Compared 6 classifiers (KNN, SVM, Naive Bayes, Decision Tree, Gradient Boosting, XGBoost) on a Seattle dataset • **XGBoost** reached **87.90% accuracy**	• Highly dependent on a single regional dataset, restricting model generalizability to highly volatile monsoon zones like India

Ref No.	Year	Authors	Paper	Research	Gaps
[4]	2025	P. Sahu, S. Kumar, R. K. Yadav, et al.	Explainable AI-Powered and Machine Learning Approach for Heavy Rainfall Prediction	• Developed a hybrid **Voting Classifier Ensemble** • Used **SHAP** and **LIME** for model transparency and feature importance tracking.	• Complex ensemble approach requires **high computational footprint** and lacks **real-time sequential multi-step timeline adjustments**.
[5]	2024	S. P. S. Ibrahim, N. V. Harika, et al	Application of Machine Learning and Data Mining Techniques for Accurate Cloud Burst Prediction	• Utilized **Fuzzy Association Rule Mining** • Handled severe class imbalances to achieve a baseline **94% accuracy**.	• The data mining rules struggle when handling datasets with significantly high spatial or temporal density variances.

Table-1

Flow Chart: Data and Logic Flow Through the System

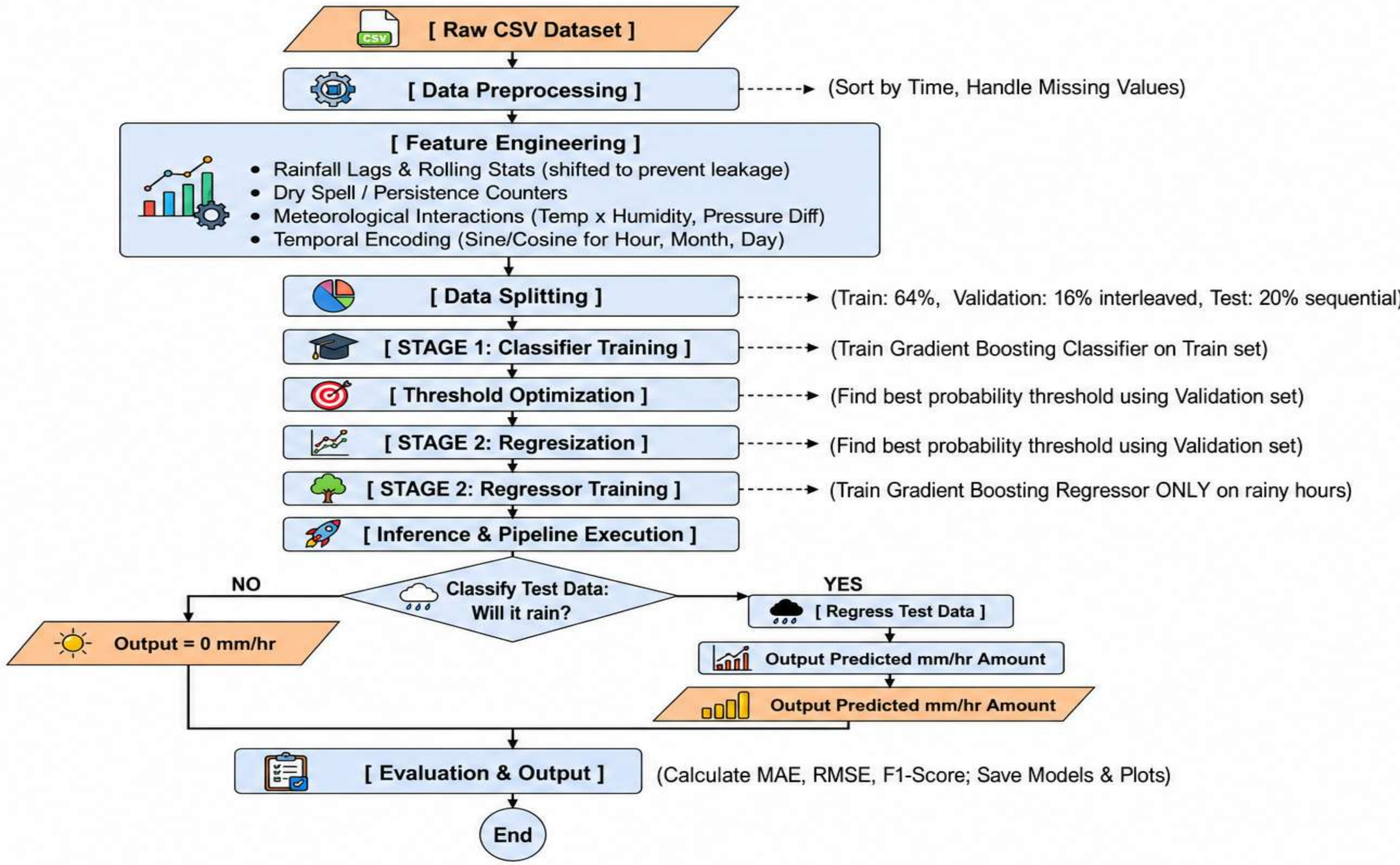


Figure-1

Block Diagram

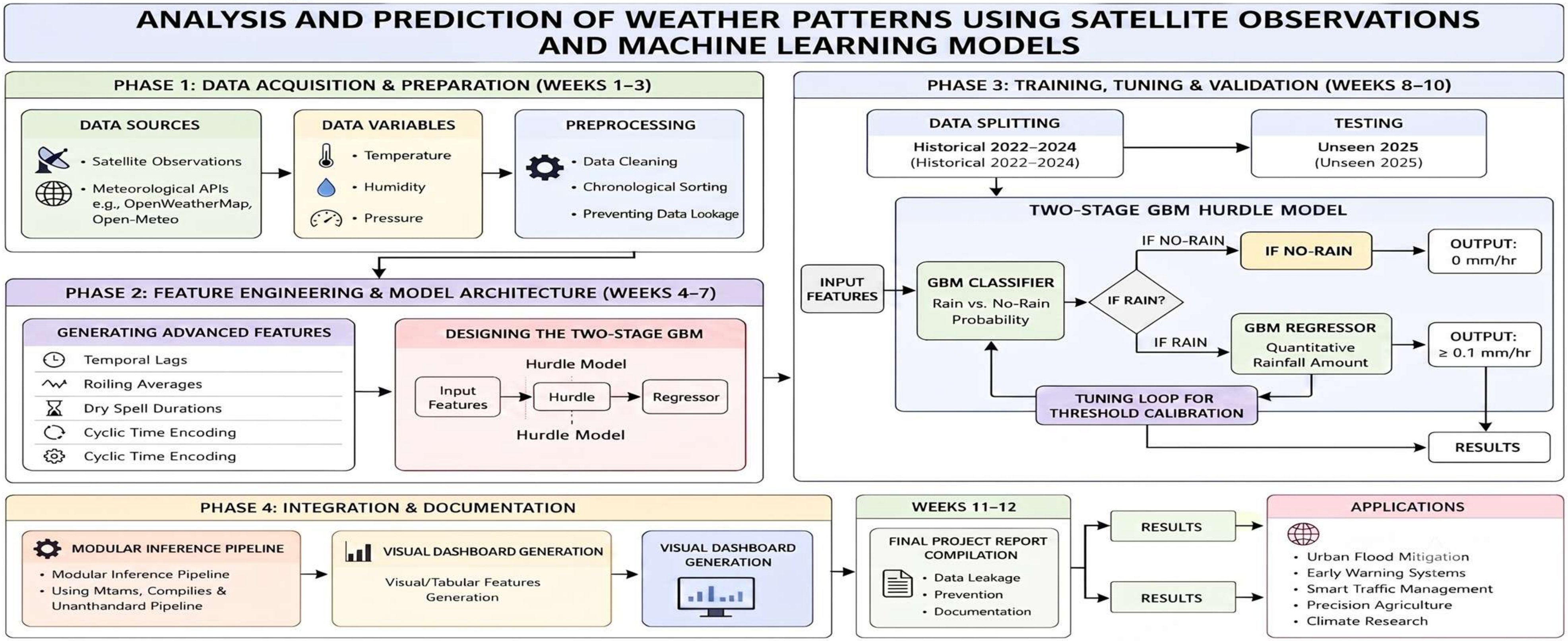


Figure-2

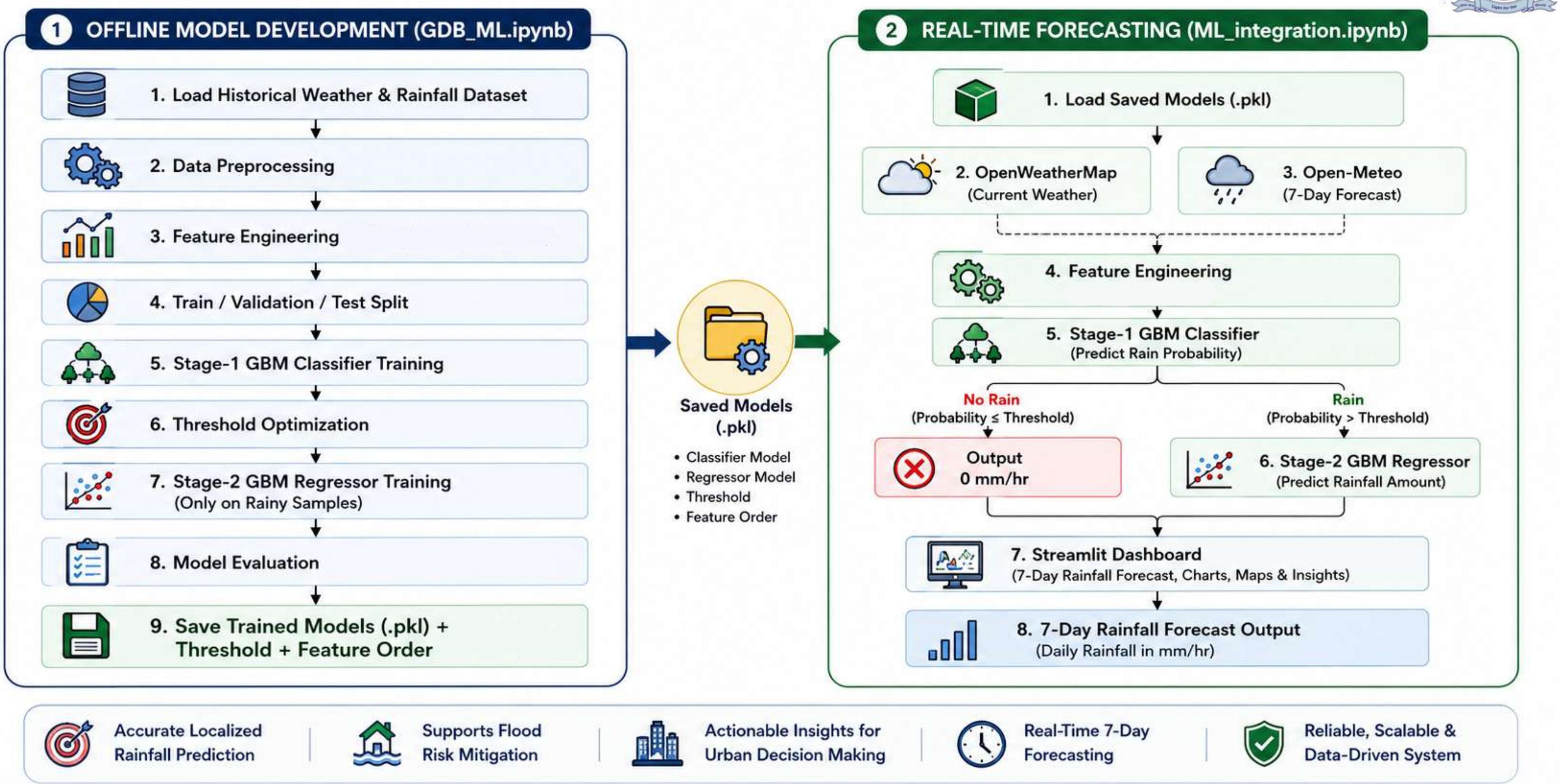
Hardware & Software Details

HARDWARE
- NodeMCU ESP 8266
- GY –SHT30-D Sensor

Output ML files
- Trained Classification Model
- Trained Regression Model
- Metadata and optimized threshold values

Software
- Streamlit
- Plotly
- Numpy & Pandas
- Scikit-learn & Xgboost
- Earthacess & Xarray
- Joblib & Request

Project Workflow: From Model Training to 7-Day Rainfall Forecast



Pictures of the Prototype Model

Figure-3 Web Dashboard

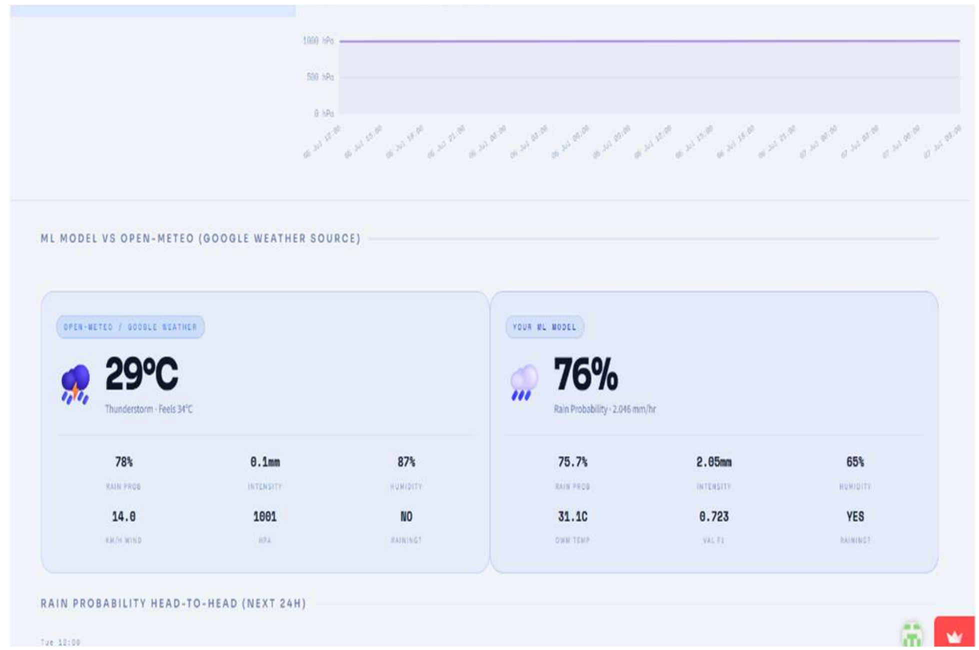


Figure-4 ML-Model Vs Open Meteo

Figure-5 Hardware Sensor Reading

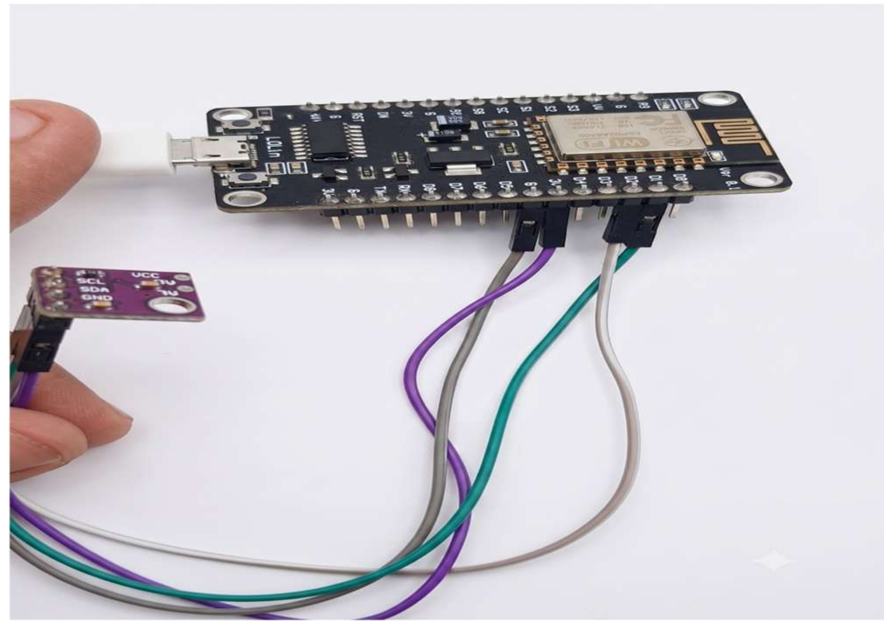


Figure-6 Hardware (ESP8266 &SHT30)

Experimental Result

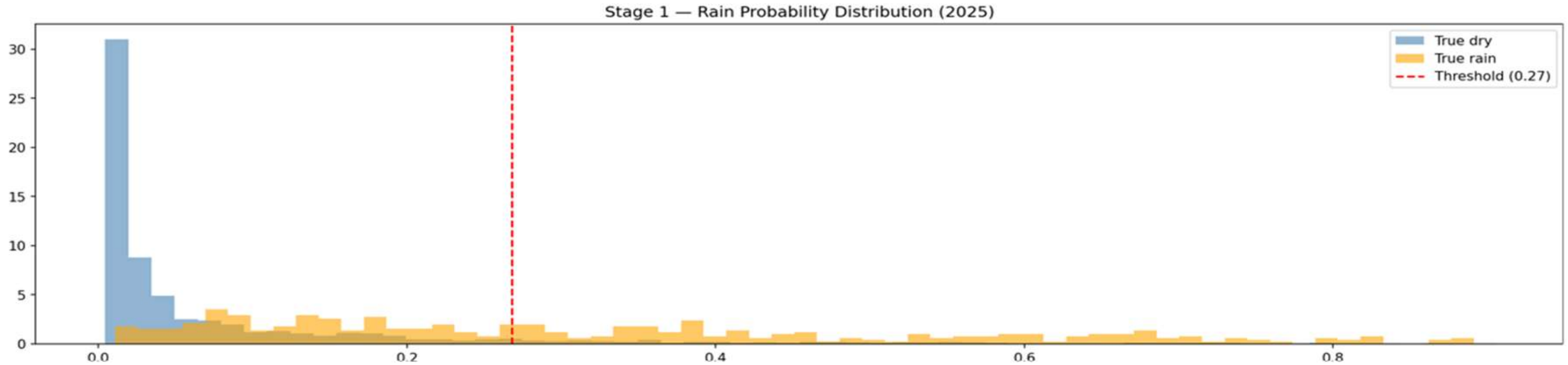


Figure-7 Stage 1 Classifier Probability Threshold

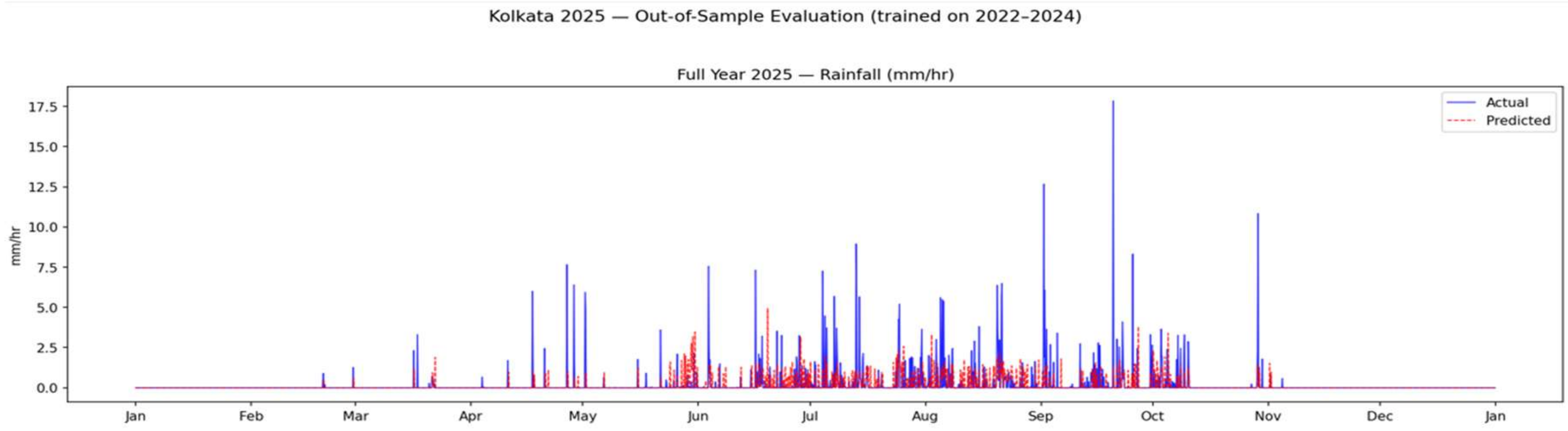


Figure-8 Full ML predication of 2025 Unseen Test data

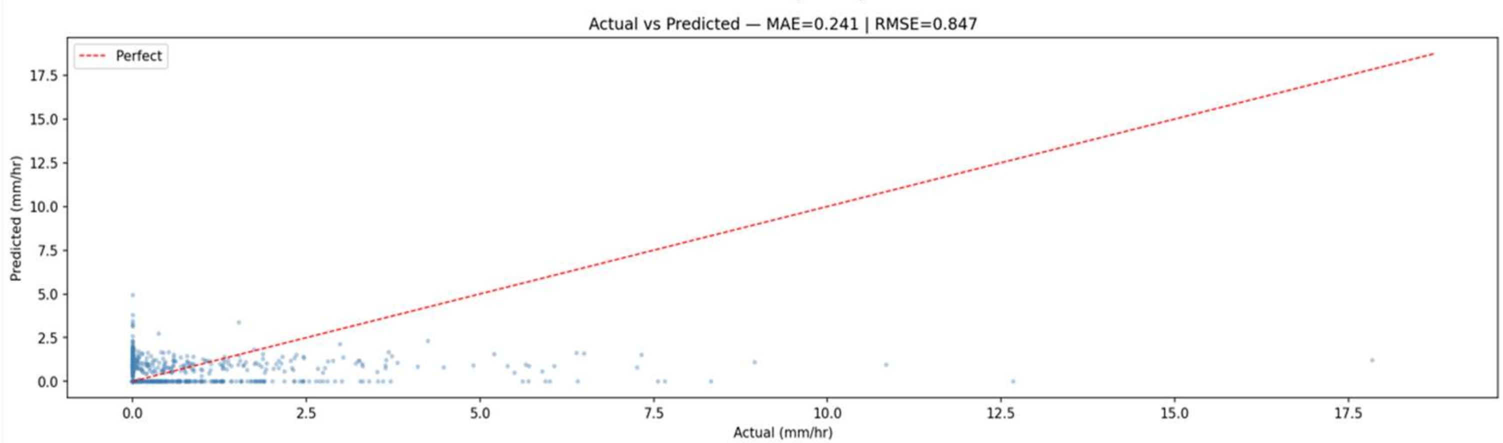


Figure-9 Actual Vs Predicated Scatter Plot

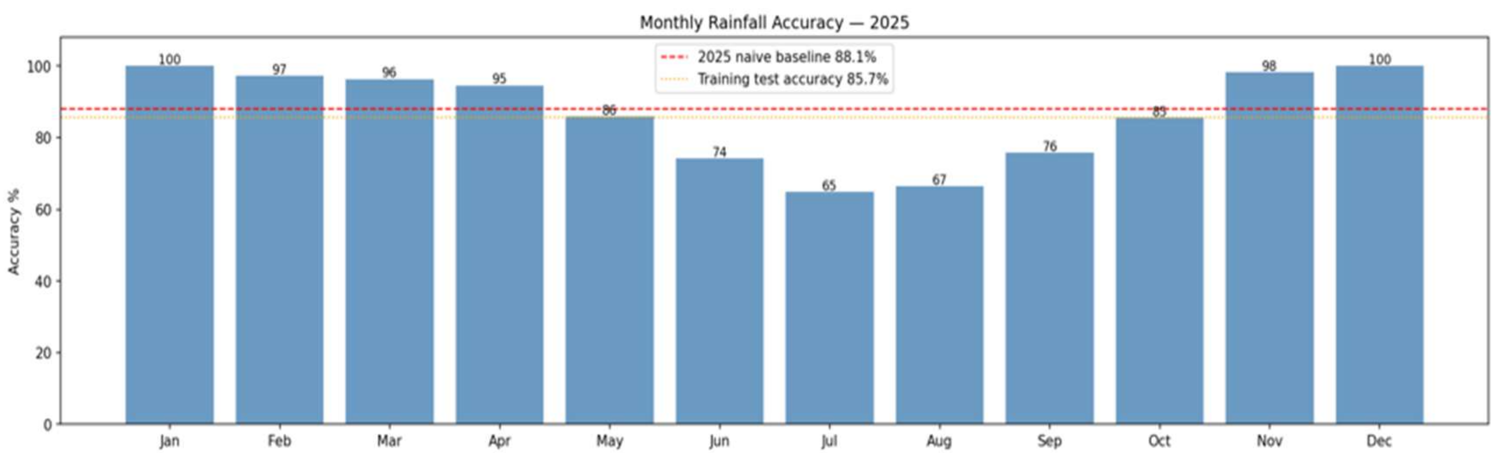


Figure-10 Monthly Accuracy Histogram

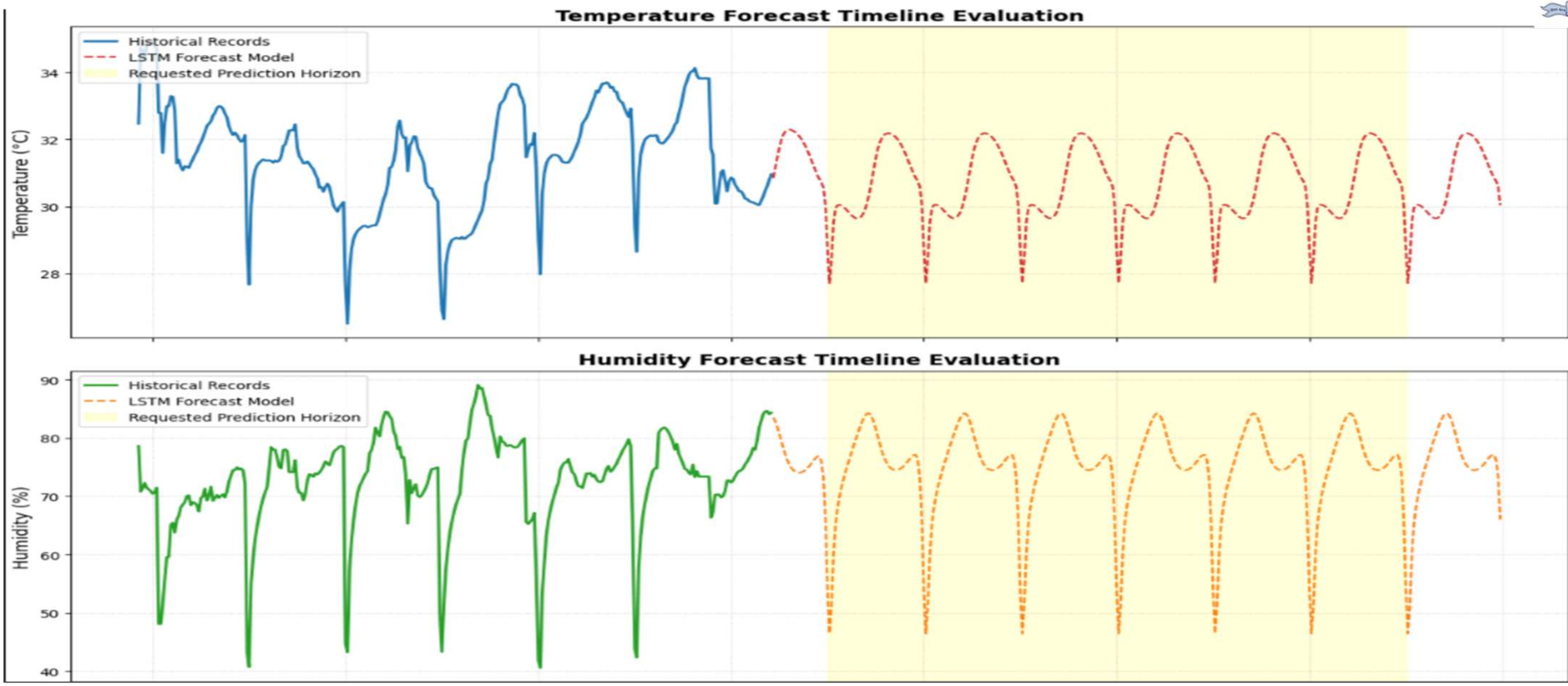


Figure-11 Hardware Statistical Performances

Analysis & Discussion of the results

ML MODEL STATS (predicting Rainfall intensity)

CATEGORY	EVALUATION METRIC	RESULT
Error Rates	MAE	0.4170 mm/hr
	RMSE	1.2902 mm/hr
Accuracy	Accuracy	81.58%
	F1- Score	0.716
Mean	Actual	4.68 mm/hr
	Predicted	0.93 mm/hr

Table-2 Training Data Stats for the (GBM) ML Model

CATEGORY	EVALUATION METRIC	RESULT
Error Rates	MAE	0.2388 mm/hr
	RMSE	0.8426 mm/hr
Accuracy	Accuracy	86.54%
	F1-Score	0.696
Mean	Actual	4.20 mm/hr
	Predicted	0.68 mm/hr

Table-3 Testing Stats of the (GBM) trained model

ML MODEL STATS (predicting Temperature and Humidity)

CATEGORY	EVALUATION METRIC	RESULT
Temperature	RMSE	0.8395 °C
Humidity	RMSE	8.8386 %

Table-4 Testing Data stats for the LSTM ML model

Future Scope

Deep Learning for Sequential Extraction: Future research must focus on optimizing advanced Deep Learning (DL) architectures, like Long Short-Term Memory (LSTM) networks or Temporal Convolutional Networks (TCNs

Hyper-Local Nowcasting: Leveraging real-time data from IoT weather sensors deployed across different municipal zones in Kolkata to develop hyper-local, real-time nowcasting (0-3 hours) for rainfall.

Extreme Event Extrapolation: Future work will focus on integrating Extreme Value Theory (EVT) to ensure the model remains robust during severe climatic anomalies.

Cost of the prototype

Category	Description	Estimated Cost
Data Acquisition	Utilization of Open-Meteo & OpenWeatherMap APIs (Free tiers)	₹0
Software & Tools	Python, Scikit-learn, Pandas, Matplotlib (Open Source)	₹0
Compute Resources	Local Workstation usage	₹0
Hardware	Sensor Microconcontroller wire etc	₹400 ~ ₹600
Misc	Printing, binding, and incidental research costs	Minimal
Total		₹700-₹800 (approx.)

Table-5 Finance

Conclusion

Dual-stage Gradient Boosting Machine (GBM) architecture—utilizing a classifier to predict rain probability and a regressor for precipitation volume—effectively overcomes the overwhelming statistical bias of "no rain" hours.

Pairing wide-area remote sensing data from **NASA's MERRA-2 and GPM instruments** with a cascading machine learning ensemble optimizes tropical weather forecasting.

Delivers accurate, localized downpour **forecasts to support flood resilience, water resource management, traffic planning**, and early disaster risk reduction in Kolkata

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- [11] Yang, Huang., George, P., Petropoulos., Qifeng, Lu., Yanfeng, Huo., Fu, Wang. “Precipitation Estimation Using FY-4B/AGRI Satellite Data Based on Random Forest.” Remote sensing,2024